

The digitalization of Formula 1: Innovations from car design to race strategy

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Abstract - Formula 1 is a prominent user of digital technologies in the world of motorsport. Digitalization has a significant impact on car design, data collection and analysis, as well as race strategy. In the realm of car design, digital simulations take the forefront, allowing designers to optimize the performance of cars even before the creation of physical prototypes. During data collection and analysis, the car's sensors constantly monitor its condition and performance, aiding teams in developing race strategies and fine-tuning the car's settings. Furthermore, digitalization enables teams to make remote car adjustments from the pit wall, providing quick responses and optimized race strategies. Digitalization in Formula 1 has allowed the teams to optimize car performance far more precisely, leading to more exciting competition.

Keywords - Formula 1, digitalization, car design, simulation, data collection, data analysis, race strategy, communication, engine control, remote-control, pit wall, race conditions, innovation, technology, sensors

I. INTRODUCTION

Formula 1 is a thrilling sport that combines the excitement of entertainment with fierce competition. It is one of the most technologically advanced and prestigious sports globally, attracting substantial investments and generating enormous revenue. At its core, F1 showcases the pinnacle of automotive engineering and innovation. The sport features teams composed of skilled drivers and dedicated engineers who work tirelessly to push the boundaries of speed, performance, and aerodynamics [4].

In recent years, Formula 1 has experienced a significant transformation through the implementation of digital technologies. The sport has embraced digitalization, leveraging innovations in various aspects ranging from car design to race strategy. This digital revolution has had a profound impact on the way Formula 1 teams operate, enabling them to extract greater performance and enhance the overall racing experience. I will now discuss various aspects, starting from car design to data collection and analysis methods, simulation, virtual testing, complex steering systems, digitized engine control, remote control from the pit wall, as well as the modern solutions that have been transferred from Formula 1 to road cars.

This research is important because it explores the profound impact of digitalization on Formula 1, a highly technologically advanced and prestigious sport. The integration of digital technologies, such as digital simulations, advanced sensors, data analysis methods, and remote-control capabilities, has revolutionized various aspects of Formula 1, from car design

and performance optimization to race strategy and communication.

The term "innovation" in this context refers to the introduction and application of novel and advanced technological solutions within Formula 1. These innovations include the use of digital simulations for optimizing car design, the implementation of advanced sensor networks for data collection and analysis, the development of remote-control capabilities from the pit wall, and the transfer of cutting-edge technologies from Formula 1 to road cars.

For automotive innovators, the role of social responsibility, (Garaí-Fodor, Popovics 2023) is exceedingly important and significant. In today's world, numerous innovations, such as solutions aimed at reducing environmental impact and enhancing traffic safety, are no longer just factors contributing to increased business results, but rather fundamental cornerstones of industry development and sustainability [4].

In today's rapidly evolving landscape, the role of consumer behavior awareness (Garaí-Fodor 2023) goes beyond mere observation, actively molding the marketing strategy of economic activities to ensure their success. This synergy of understanding consumer behavior intersects with innovative methodologies propelled by the strides in digitalization, as exemplified by the transformative impact of technological solutions within the realm of Formula 1 [3].

II. THE ROLE OF DIGITALIZATION IN FORMULA 1

A. Car design

Digitalization has a significant impact on the design of Formula 1 cars. The car design process is based on digital simulations, allowing designers to optimize the performance of cars even prior creating physical prototypes.

Designers utilize software and modelling tools to simulate the car's aerodynamics, suspension, engine performance, and other critical factors. This enables designers to experiment with different settings and modifications virtually and evaluate their impact on the car's performance.

The digital design process not only saves time but also enables faster iteration and decision-making. Designers can easily modify the car's parameters and immediately see the results in simulations. This allows for faster detection and correction of design errors and enhances the efficiency of the design process.

The design process is like a continuous loop of improvement and refinement. It all starts with an initial idea or concept for the car's aerodynamics. This concept is then translated into multiple variations using computer-aided design (CAD) software called NX [5].

Once they have the 3D models of these variations, the teams move on to the virtual testing phase. They use a process called computational fluid dynamics (CFD) to simulate and evaluate how the air flows around the car. This helps them understand the aerodynamic performance and make necessary adjustments [15] [16].

Based on the results from the CFD simulations, the team selects the most promising design and moves on to the next step. They create a 60% scale model of the car and put it through testing in a wind tunnel. By comparing the wind tunnel results with the CFD simulations, they can validate and fine-tune their aerodynamic understanding [15].

After analyzing the data from the wind tunnel tests, the teams decide on the final design for the full-sized car. They consider the information gathered from the previous tests to make an informed choice. This selected design is then translated into the NX software, where virtual tools from the Siemens Xcelerator platform are used for a process called finite element analysis (FEA). FEA helps ensure that the car's components are structurally sound and capable of withstanding the demanding conditions on the racetrack [5].

Once the virtual validation is complete, the team proceeds to the manufacturing stage. The parts for the full-sized car are produced based on the finalized design. These parts then undergo rigorous testing to verify if they meet the expectations set during the virtual validation process. If the tests are successful and the parts perform as expected, they are integrated into the Formula 1 car [5] [15].

The first real-world evaluation of the car happens during practice sessions. This is where the team gets to see how the car and its components performs on the track. They can compare the real-world results to the previous simulations and tests conducted virtually. If everything aligns and the car shows good performance, they continue with racing. However, if any issues or discrepancies arise, they go back to the iterative loop, analyzing the data, making improvements, and repeating the process throughout the season [5] [15].

B. Data collection and analysis methods

Cars are equipped with modern technology that utilizes sensors and data collection systems to monitor the car's condition and performance. The sensors on cars continuously gather data on speed, steering, braking, aerodynamic forces, engine performance, suspension settings, and many other parameters. The data is digitally recorded and transmitted to teams through the telemetry system [15] [16].

Modern Formula 1 race cars are highly advanced machines with extensive sensor networks and intricate wiring. These sensors collect a massive amount of data, enabling teams to analyze and extract valuable insights. With this wealth of information, teams can gain a competitive advantage by making informed decisions and optimizing the car's performance on the track [13] [16].

In a Formula 1 car, multiple interconnected systems like the engine, exhaust, gearbox, differential, and aerodynamics work together. The Engine Control Unit (ECU) acts as the central hub (will be discussed in more detail later), with the Standard ECU (SECU) serving as a powerful computer for data storage and transmission. The car also features a Power Control Module (PCM), a Data Acquisition Main Control Unit (MCU), the Accident Data Recorder (ADR), which collects data and gives an instant alert if an accident occurs,

it is a "black box". Furthermore, the most recognized and one of the most important high-end technology parts of the car, the steering wheel, which allows the driver to get feedback from the car and enables them to change various settings. These components enable efficient communication and data management within the car [13].

There are hundreds of sensors at various points in the car. Sensors fall into three different categories, which can be magnetic, optical, and even laser driven [13] [15] [16]:

- 1) *Instrumental sensors (pressure, fuel flow)*
- 2) *Monitoring sensors that send data channels about the status of the car's systems*
- 3) *Control sensors that convert driver inputs into car outputs.*

The sensors and components communicate with each other through an onboard network. The data collected is logged onto an onboard server and then encrypted before being transmitted via radio frequency through an antenna on the car. This communication happens almost instantly. Formula 1 Management (FOM) created a standardized communication network. It utilizes fiber links and shared access points to provide encrypted communication between the car and the team's garage. Real-time data is transmitted in small packets, while during a pit stop, the car can emit a burst of data using microwave technology when in range [13].

Throughout a race weekend, each Formula 1 car generates more than 1 terabyte of data, including video and additional information. This amount significantly increases once teams process the data further during or after the event. At the team's factory, a massive volume of data is also generated and analyzed. Transferring the data from the track to the factories is a challenging task due to the enormous amount of data generated at the circuit. During European races, the data can be transmitted back to the factories in just 10 milliseconds, almost instantaneously. However, for races in distant locations like Australia or Japan, which are thousands of miles away, the data takes approximately 300 milliseconds to reach its destination. This slight delay is still remarkably quick, but it is longer compared to races held closer to the teams' home bases [10] [16].

There is limited testing time, so it's crucial to make the most of every opportunity. The same applies to the data gathered from the car. It needs to be accurate and collected effectively to extract valuable insights. When the data is collected, it is synchronized to ensure precise time alignment across all sensors. It is then encrypted and transmitted back to the team's garage using a telemetry system shared among all teams. Drivers are eager to improve their performance, so they spend time reviewing the data after each session. They compare themselves to their teammates, analyze braking data, and examine corner speeds. The engineering groups also dive into the data to uncover valuable insights. With vast amounts of data available, prioritizing the information for analysis becomes an engineering challenge. The goal is to quickly identify the relevant data and learn as much as possible to enhance the driver's performance and make the car faster for future races [10] [16].

C. Simulators, simulation and virtual testing

Digitalization and simulation are closely interconnected in the world of Formula 1, particularly using simulators by teams. Simulations are carried out using digital models that

allow for the virtual reproduction of real racing conditions and the prediction of car performance.

A dedicated department is responsible for the design, development, and maintenance of Formula 1 simulators. These simulators are built using an actual monocoque, complete with a driver's seat and a six-point harness. The monocoque is mounted on a hydraulic platform with multiple legs, which accurately replicates the dynamic effects experienced by the car on the track. The simulator moves in response to real data from the track, providing an immersive and realistic driving experience for the drivers and engineers. The simulator creates a "digital twin" of the physical car [15].

The simulator has become a vital tool in Formula 1 due to limited track testing. Teams can conduct a much larger number of tests and simulations compared to physical car testing. This not only enhances performance but also proves to be environmentally friendly and cost-effective. The simulator is a powerful and efficient tool for gathering intelligence and optimizing strategies [12] [15].

The simulator has two main tools [12]:

1) *The Race Planner uses visualizations to map out different scenarios for each driver, helping to guide discussions and strategies.*

2) *The Monte Carlo Model runs millions of simulations, combining random factors to determine the probability of various outcomes. This information is crucial for the race strategy team, as it allows them to develop plans to influence the likelihood of certain outcomes and adapt to unexpected situations during the race.*

Simulators in Formula 1 leverage algorithms and machine learning to transform vast amounts of data into meaningful insights. Each aspect of the car is modeled individually, allowing development drivers to provide feedback on different setups and tracks. This feedback is then used to enhance the simulator, improving its realism and effectiveness. Simulators are typically operational for several days each week, running extensive simulations equivalent to two full race distances per day. This generates a significant amount of data, around 100 GB per week, which is carefully analyzed by the team. The advantage of simulators lies in the ability to work with clean and precise data, free from external variables like other cars on the track. This ensures a comprehensive understanding of the car's behavior and performance [12].

Simulators use data visualizations to predict race outcomes and assist teams in planning for various scenarios. By analyzing millions of data points, teams can forecast lap performance and strategize different car configurations accordingly. Plans are discussed, practiced, and prepared for execution, ensuring readiness for different situations. Simulators play a crucial role in the continuous data exchange between the track and the factory throughout the race weekend. Testing data from the track is sent to the factory, allowing engineers to replicate and address any issues encountered. Solutions are then communicated back to the race engineers, creating a collaborative approach that covers all aspects. Data serves as the vital link connecting everyone involved, working together towards the shared objective of achieving victories [12] [15].

D. Steering wheel

The steering wheel of Formula 1 cars is a complex and intricate structure that includes numerous adjustment options and control elements. However, the development of the steering system is not directly linked to digital technologies, but rather connected to the improvement and optimization of the cars' aerodynamic and mechanical properties. The driver can adjust the car's systems using the buttons on the steering wheel, including brake bias, brake pressure, differential rotational speed control, and engine mapping [9] [11].

Let's see what can be done with the steering wheel besides steering [9]:

1) *Strategy Settings: Every race is different, and each has its own characteristics. Engineers, along with the drivers, develop numerous different race strategies for each race. They prepare for every possible scenario, such as malfunctions, rain, or unexpected temperature changes. Each team has its strategic solutions that can be adjusted using the rotary knobs on the steering wheel.*

2) *MGU-K Settings: In the world of Formula 1, the powertrain of an F1 car is an extremely complex and sophisticated system, especially due to the hybrid technology introduced in recent years. This complexity brings the possibility of various types of malfunctions. To address this, engineers of an F1 team have developed special engine calibration maps that allow the driver to manage and mitigate certain faults while preventing others. To provide flexibility and control to the driver, a rotary knob has been placed on the steering wheel, enabling adjustments to specific settings related to the car's power unit, particularly the Motor Generator Unit-Kinetic (MGU-K). These settings are designed to cater to different scenarios and conditions that the driver may encounter during the race. Examples of such settings include adjustments tailored for wet weather conditions, where performance needs to be limited to maintain traction and prevent wheelspin. During qualifying sessions, where every fraction of a second counts, there are special performance settings that unleash the car's maximum potential. Throughout the actual race, various performance settings come into play. The race performance setting focuses on striking a balance between performance and fuel efficiency, allowing the driver to maintain a competitive pace while managing fuel consumption over the entire race distance. Additionally, there are settings specific to low fuel consumption situations. In cases where certain malfunctions or issues arise within the powertrain, there are even specific performance settings that mitigate the impact of such faults.*

3) *Menu: The driver has access to a comprehensive menu system that grants them control over nearly all of the car's intricate systems. This menu acts as a centralized hub, allowing the driver to navigate through various pages of telemetry and make adjustments to optimize the car's performance. The menu provides a wealth of information and data about the car's condition and performance in real-time. It displays telemetry such as speed, engine revs, tire temperatures, fuel levels, brake temperatures, and much more. Beyond simply monitoring the telemetry, the menu also empowers the driver to make adjustments and fine-tune the car's settings.*

4) *Differential Balance*: A differential balance can be controlled with a single scroll button or a set of scroll buttons. This allows the driver to make small adjustments to the settings. During the race, drivers can shift the balance left or right to ensure the appropriate balance in every corner. This is also true for brake balance.

5) *Energy Recovery*: In certain cases, referred to as "Harvest," drivers have the opportunity to recover a portion of the energy generated during braking and store it in the car's hybrid system for later use.

6) *Race Start*: The Race Start button sets a specific speed limiter to hopefully achieve a perfect off-the-line launch at the first corner. This is similar to launch control but relies much less on the car's electronics.

7) *Pit-Lane Speed*: Given that there are people working in the pit lane, it is necessary to limit the speed of the cars so that they do not arrive at speeds of 200 miles per hour. That's why there is a button that restricts the car's speed when entering the pit lane, so the driver doesn't have to worry about maintaining the designated speed. They just need to press this button before reaching the pit lane line.

8) *Overtake*: The Overtake button, also known as the push to pass button, momentarily increases the car's performance to overtake the car ahead.

9) *DRS*: The DRS stands for Drag Reduction System, which is an expression of active aerodynamics. In the case of Formula 1, it refers to a semi-active rear wing. The driver can activate the DRS button when they are within one second of the car ahead in designated DRS zones on the track. Upon pressing the button, the central part of the rear wing opens, reducing drag. In the open position, the DRS provides the driver with approximately 6-10 miles per hour of additional speed by the end of the zone, although this may vary depending on the track.

10) *Some other important buttons*: Pit Confirm, Radio, Drink and other secret, pilot- and team-specific setting buttons.

E. Engine control

While the digitization of the engine control was previously mentioned, I would like to delve deeper into the details in this section.

The engine control in Formula 1 cars is also digitalized. Advanced electronic control systems are utilized for the engines in modern Formula 1 cars, enabling the optimization of engine performance and driving experience. The Engine Control Unit (ECU) monitors and regulates various parameters of the car's engine. Using sensors, the ECU continuously collects information about the engine's condition, such as rpm, temperature, pressure, and fuel supply. This acts as the brain of the engine, controlling and managing various systems of the car. It is composed of electronic and software components, such as microprocessors, sensors, and actuators, which work together to regular and oversee the engine's functions, such as fuel injection and ignition [7] [15] [16].

The ECU (Electronic Control Unit) records various information about the car's performance and systems. Teams then use this information to optimize the car's performance,

modify the engine and other systems, and make strategic decisions during the race [7] [15] [16].

The ECU has revolutionized the management of teams' engines and other systems in the car. It has enabled more precise control over power delivery and telemetry, allowing teams to make real-time adjustments during races. It has also played a key role in the advancement of Formula 1 technology, pushing teams towards continuous development and innovation to gain a competitive edge. In the future, technological advancements are expected to further enhance the performance of cars and provide even greater control for teams over their engines and systems [7].

F. Remote control from the pit wall

The box strategies, tire changes, fuel and tire management, and adaptability to unforeseen circumstances are all crucial factors that can determine the outcome of a race. The driver alone is unable to make these decisions while racing. This is where the pit wall comes into play. The pit wall, located opposite the team's garage in the pit lane, serves as the "nerve center" of a Formula 1 team during a race. It is manned by a group of key personnel who make critical decisions regarding the race management of the team's cars. They provide realtime updates to the driver regarding their position, the positions of rivals, and the race situation. Furthermore, the pit wall functions as a control center, equipped with advanced technology and monitors displaying telemetry data, race information, and timing data. This allows the team to analyze the performance of their cars, track conditions, and other crucial factors. They can also receive important updates from race control, such as weather forecasts and track condition notifications, which assist in fine-tuning the race strategy for optimal results. The pit wall is connected to various communication channels, enabling continuous radio communication with team members, including the drivers on the track. This facilitates seamless coordination and information exchange during the race. Moreover, the pit wall maintains contact with the FIA race director, ensuring compliance with rules, regulations, and technical aspects of the race [1] [14].

The teams in Formula 1 have advanced technology that allows them to remotely intervene and adjust the car's settings and parameters from the pit wall. These control systems provide teams with the ability to modify various aspects of the car's performance, such as its aerodynamics, suspension, and fuel mixture regulation. Through digitalization and remote-control capabilities, teams can swiftly respond to changes in race conditions and fine-tune the car's performance for optimal results. Engineers and technicians stationed at the pit wall have access to real-time data from the car, which they continuously monitor and analyze. This data allows them to make informed decisions and intervene in the car's settings without physically being on the track. This remote-control capability provides teams with greater flexibility in managing race strategy and maximizing the car's performance throughout race weekends. By adjusting based on the data and observations gathered from the pit wall, teams can optimize the car's settings to adapt to evolving track conditions, tire wear, and other variables that may affect performance [14] [15].

G. Transfer the modern solutions from Formula 1

Formula 1 technology goes beyond racing itself. The technological innovations that can be found here have a great impact on other areas as well.

Here are some examples of solutions originated from Formula 1 that made their way to be used in road cars [2] [8]:

1) *Double Overhead Camshaft engines*: The camshaft in an engine controls the opening and closing of the intake and exhaust valves. With a double camshaft, car manufacturers can incorporate multiple valves per cylinder and strategically position them for improved efficiency. This technology is commonly used in road cars. However, Formula 1 teams are exploring alternative methods such as electromagnetic and electrohydraulic actuators to operate the valves, aiming for further advancements in performance.

2) *Carbon fibre*: McLaren pioneered the use of carbon fiber in road cars by introducing the world's first series production car with a carbon fiber chassis, the Mercedes-Benz SLR McLaren. This breakthrough led to the adoption of carbon fiber in various high-end road cars from brands like Porsche, BMW, and Aston Martin. The expertise developed by Formula 1 teams in carbon fiber technology has also been utilized in other sports such as sailing, bobsleigh, and cycling, demonstrating its widespread application beyond the realm of racing.

3) *Paddle shifter*: Invented in the 19th century, made their way into Formula 1 in the 1989 season with Ferrari's introduction of a semi-automatic gearbox. This innovative system utilized two paddles mounted behind the steering wheel, allowing drivers to shift gears seamlessly. The success of paddle shifters in F1 led to their adoption in Ferrari's road cars, with the technology appearing in the F355 just eight years later.

4) *Steering-wheel buttons*: The buttons and controls we find in our cars today, like volume control, radio station selection, driver's display, and cruise control, can be traced back to Formula 1. This trend began in the 1970s but gained momentum in the '80s and '90s as F1 cars embraced advanced technologies. To ensure convenience and safety, drivers wanted these controls within easy reach, so they were integrated into the steering wheel itself.

5) *Diamond-like cylinder coatings*: F1 teams employ a diamond-like coating on their cylinder walls to minimize friction. This process involves depositing an extremely hard carbon-based material through vapor deposition. The resulting coating exhibits properties that combine the hardness of diamond with the lubricating qualities of graphite. This technology has made its way into high-end production cars, with the Ferrari 458, introduced in 2010, being among the first vehicles to feature this advancement.

6) *Adaptive suspension*: Formula 1 pioneered the technology of adjustable suspension, which allows cars to modify their suspension settings with the press of a button. This innovation first emerged in the mid-1980s, with early applications focusing on maintaining a consistent ride height to optimize downforce. This breakthrough has since influenced and inspired the development of adjustable suspension systems in various road cars.

7) *Hybrid powertrains*: Since the 2014 season, all Formula 1 cars are equipped with hybrid drivetrains that incorporate two types of energy recovery: the MGU-K (Motor Generator Unit - Kinetic) and the MGU-H (Motor Generator

Unit - Heat). The MGU-K captures braking energy, while the MGU-H harnesses heat from the turbocharger. This recovered power is stored in a lithium-ion battery. Compared to the previous V8 engines used until 2013, the current V6 turbo-hybrid power units offer 20% more power while emitting 26% less CO₂. The thermal efficiency of the V8 engines peaked at 29%, whereas the introduction of V6 turbo-hybrids in 2014 significantly increased it to around 40%, and it now surpasses 50%. The turbo-hybrid power units are also faster than their V8 counterparts. The success and efficiency of this technology have led to its adoption in road cars. For example, the Mercedes-AMG Project One hypercar features a 1.6-liter turbo-hybrid power unit derived from Formula 1, and Aston Martin's Valkyrie, developed with input from Red Bull's Chief Technical Officer Adrian Newey, incorporates an energy recovery system. Additionally, Mercedes utilizes hybrid technology inspired by its Formula 1 car in models like the S-Class, while paddle shift gearboxes, originally developed in Formula 1 in the late 80s and early 90s, are now commonly found in road cars.

It is important to note that the transfer process can take time as road cars involve various other factors to consider, such as regulatory requirements, safety standards, and operational conditions. Moreover, during the transfer, certain technologies or features may undergo modifications or limitations for their use in road cars.

III. SUMMARY

Digitalization has revolutionized Formula 1 in multiple ways, bringing significant advancements across various aspects of the sport. Car design has benefited immensely from digital simulations, allowing teams to optimize performance and aerodynamics without relying solely on physical prototypes. The integration of advanced sensors has greatly enhanced data collection and analysis, providing teams with valuable insights into the car's condition and performance. This data is utilized to develop effective strategies and finetune settings for optimal results. Furthermore, digital technologies have enabled real-time communication and remote adjustments between the pit wall and the car, leading to swift responses and optimized race strategies. The impact of digitalization extends beyond Formula 1, as many of the innovations pioneered in the sport have influenced advancements in road driving, leading to the implementation of innovative solutions in production vehicles. Overall, digitalization has elevated the level of competition in Formula 1 and has had a significant impact on the automotive industry.

The following table summarizes the key digital solutions that have emerged over the years and are still in use in Formula 1 today:

<i>Solution</i>	<i>Utilization</i>	<i>Appearance</i>
Sensors	Data collection	1970s
Simulators	Simulation	1980s
3D CAD systems	Car design	Mid 1980s
Power Control Module (PCM)	Engine control	Mid 1980s
Monte Carlo Model	Simulation, Race strategy	1990s

CFD simulations	Car design	Mid 1990s
Data Acquisition Main Control Unit (MCU)	Data collection	Mid 1990s
Digital steering wheel	Steering	Mid 1990s
Strategy settings on steering wheel	Race strategy	Mid 1990s
Accident Data Recorder (ADR)	Data collection	Late 1990s
Engine Control Unit (ECU)	Engine control	Late 1990s
Formula One Management Telemetry System	Real time data connection	2008
Energy Recovery	Engine control	2009
Drag Reduction System (DRS)	Overtaking, reduction of air resistance	2011
MGU-K setting on steering wheel	Engine control	2014
Race Start button	Optimalization	2015

Theoretical contributions of this research encompass the digital transformation in motorsports, illustrating how Formula 1 has embraced digital technologies to enhance car design, data collection, analysis, and race strategy. This demonstrates the adaptability and potential benefits of integrating digital solutions into sports. Furthermore, the study explores performance optimization through digital simulations and data analysis methods. These contributions shed light on how these methods aid teams in fine-tuning car settings, developing effective strategies, and swiftly responding to changing race conditions.

Practical contributions of this research include the advantageous race strategy, where digital technologies enable teams to intervene in real-time and optimize race strategies from the pit wall, thereby improving performance and achieving better results. Furthermore, the technology transfer, where the exploration of technology transfer from Formula 1 to road cars showcases practical applications and highlights the benefits of innovations originating from the high-performance racing environment.

The limitations of the research are, on the one hand, the concentration on digitization. The research mainly focuses on the role of digitization in Formula 1, another important factor contributing to the success and innovation of the sport, which may be overlooked. On the other hand, the coverage is not complete. Although the research provides an in-depth analysis of the impact of digitization on various areas of Formula 1, it does not cover all possible technological developments and impacts.

In summary, this research highlights how digitalization has transformed Formula 1 by revolutionizing car design, data collection and analysis, race strategy, and communication. The innovative technologies developed within Formula 1 also have the potential to influence and shape advancements in road car technology.

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